

**Is a standardized global nutritional index an effective tool for
measuring and addressing malnutrition across countries with diverse
socio-economic and cultural contexts?**

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1. Abstract

Malnutrition remains a major global challenge, affecting both low- and high-income countries through undernutrition, micronutrient deficiencies, and rising obesity. Efforts to track and address these issues often rely on standardized global nutritional indices, such as the Global Hunger Index (GHI) and measures from the State of Food Security and Nutrition in the World (SOFI) report. These tools provide valuable comparability across nations, highlight global trends, and support advocacy by showing the scale of the problem—for instance, recent data estimate that about 2.6 billion people cannot afford a healthy diet . However, evidence shows that a single global index cannot fully capture the diversity of socio-economic conditions, dietary patterns, and health risks across countries. In India, for example, debates around the GHI highlight that some global indicators do not directly measure hunger or reflect local realities. Population-specific health thresholds, such as Asian BMI cut-offs, further show the need for context-sensitive metrics. This paper argues that while a standardized index is useful for global monitoring and accountability, effective solutions require a hybrid framework: a global core set of indicators complemented by local adaptations that reflect cultural diets, sanitation, and intra-country inequalities.

Keywords: malnutrition, nutritional index, global health, affordability of healthy diet, Global Hunger Index, SOFI, India, socio-economic diversity, BMI cut-offs, multidimensional poverty.

2. Introduction

Malnutrition is one of the most serious global health and development problems today. It takes different forms: undernutrition (low weight, wasting, stunting), hidden hunger (lack of vitamins and minerals), and overnutrition (overweight and obesity). Together, these problems are often called the “triple burden of malnutrition.” According to the FAO, IFAD, UNICEF, WFP, and WHO (2024), about 2.6 billion people in the world were unable to afford a healthy diet in 2022, while millions also

suffer from obesity and diet-related non-communicable diseases such as diabetes and heart disease.¹ Because the problem is global, international organizations and researchers often use standardized indices to measure and compare malnutrition across countries². Some important tools include the Global Hunger Index (GHI), the FAO Food Insecurity Experience Scale (FIES), and child nutrition indicators such as stunting, wasting, and underweight reported by WHO and UNICEF. These indices provide one number or score that allows easy cross-country comparison. They are widely used in reports, policy discussions, and global monitoring, especially for Sustainable Development Goal 2 – Zero Hunger. For example, the GHI ranks countries each year and has been important in drawing attention to hunger and nutrition issues in South Asia and Sub-Saharan Africa³.

However, the use of a single global index is debated. One problem is that averages can hide inequalities inside a country. In India, for instance, nutrition outcomes differ widely between states, districts, urban and rural areas, and between social groups. Critics of the GHI have also argued that it does not directly measure hunger, and that some of its indicators (such as child mortality) may reflect broader health conditions rather than food access (Indian Council of Medical Research, “Global Hunger Index does not really measure hunger — An Indian perspective,”⁴ Another challenge is cultural and dietary diversity. What counts as a “healthy diet” in one region may look very different

¹ Fu, H., Moncayo, J. R., Bai, Y., Conti, V., Rissanen, M., & Cafiero, C. (2025, August 6). *Can everyone afford to eat healthy? New data show progress, but not everywhere*. World Bank Blogs. <https://blogs.worldbank.org/en/opendata/can-everyone-afford-to-eat-healthy--new-data-show-progress--but-not-everywhere>.

² FAO et al., *State of Food Security and Nutrition in the World 2024*, <https://doi.org/10.4060/cc8520en>). This shows that malnutrition is not only a problem of poverty but also of poor diet quality and rising inequalities.

³ World Health Organization. (2025). *Global Hunger Index (GHI)*. Nutrition Landscape Information System (NLiS). <https://www.ifpri.org/topic/global-hunger-index>

⁴ Press Information Bureau. (2021, December 3). *Global Hunger Index 2021* [Press release]. Ministry of Women and Child Development, Government of India. <https://www.pib.gov.in/PressReleasePage.aspx?PRID=1777719>).

in another. For example, the EAT-Lancet reference diet, designed as a global model, was found to be unaffordable for 1.6 billion people worldwide and often does not match local eating patterns.⁵

This brings us to the central research question of this paper: ***Can a standardized global nutritional index truly capture the diverse realities of malnutrition across different socio-economic and cultural contexts?***

The objective of this paper is threefold. First, it aims to evaluate the effectiveness of a standardized global index for measuring malnutrition and guiding international policy. Second, it seeks to analyze the challenges that arise when such indices are applied in countries with very different social, cultural, and economic conditions. Finally, it will suggest improvements to make these tools more meaningful, including the possibility of hybrid approaches that combine a global “core” with locally adapted indicators.

2.1 What is a Nutritional Index?

A nutritional index is a composite measure that brings several pieces of nutrition evidence together into one score so that countries and regions can be compared and progress can be tracked over time. The main aim is to make it easier for governments, researchers, and international agencies to measure progress, compare countries, and identify gaps in nutrition outcomes. In simple words, instead of looking at many different numbers separately such as calorie intake, diet diversity, or child stunting, a nutritional index combines them into one measure. This helps to give a broad picture of how a population is doing in terms of food security and health. In practice, it pulls data from four main areas: (1) food access and affordability (for example, the FAO “Cost of a Healthy Diet” estimates the cheapest locally available foods that meet healthy diet standards), (2) experience of food insecurity (the FAO Food Insecurity Experience Scale asks eight standard questions about

⁵ Hirvonen et al., 2020, *Lancet Global Health*, [https://doi.org/10.1016/S2214-109X\(19\)30447-4](https://doi.org/10.1016/S2214-109X(19)30447-4).

people’s difficulties in getting enough safe and nutritious food), (3) diet quality and **diversity** (indicators such as Minimum Dietary Diversity for Women and the minimum acceptable diet for young children), and (4) health outcomes (child stunting, wasting, underweight/overweight, sometimes anemia), which show the biological results of poor diets and disease⁶. These components are selected, put on a common scale, and combined with weights to create a single, easy-to-read number—an approach used in platforms like the Global Nutrition Report and in UN monitoring systems⁷.

Different organizations use slightly different designs, but most indices have some common components. First is food access and affordability, such as the FAO’s Cost and Affordability of a Healthy Diet (CoAHD), which estimates the minimum daily cost of eating a diet that meets WHO standards for energy and nutrients.⁸ Second is the experience of food insecurity, measured by surveys like the Food Insecurity Experience Scale (FIES), which asks households about their struggles in getting safe and adequate food. Third are diet quality and dietary diversity indicators, including measures such as Minimum Dietary Diversity for Women (MDD-W) and the Minimum Acceptable Diet (MAD) for children. These indicators show whether people eat a balanced range of foods, not just calories. Finally, health outcomes like child stunting, wasting, underweight, obesity, and anemia are often included, as they reflect the long-term biological impact of poor diets, poor sanitation, or disease environments.

⁶ Food and Agriculture Organization of the United Nations. (n.d.). Voices of the Hungry: Food Insecurity Experience Scale (FIES). FAO. <https://www.fao.org/in-action/voices-of-the-hungry/fies/en/>

⁷ (Food and Agriculture Organization of the United Nations, *The state of food security and nutrition in the world 2024: Cost and affordability of a healthy diet*. FAO. <https://openknowledge.fao.org/server/api/core/bitstreams/39dbc6d1-58eb-4aac-bd8a-47a8a2c07c67/content/state-food-security-and-nutrition-2024/cost-affordability-healthy-diet.html>

⁸ FAO, 2024, *SOFI Report* – <https://doi.org/10.4060/cc8520en>).

When all of these measures are collected, they are standardized, weighted, and combined into one score—sometimes a number, sometimes a ranking. For example, the Global Nutrition Report publishes country-level nutrition profiles that merge diet data, food affordability, and health outcomes, while the Global Hunger Index ranks countries annually using a combination of child undernutrition, child mortality, and undernourishment⁹. The strength of such indices is that they allow clear comparisons across countries and help track progress over time in a way that is easy to communicate to policymakers and the public.

However, the weakness is that they can be too broad. A single score may hide important local differences. For example, in India, nutrition outcomes vary sharply between urban and rural areas, and between states such as Kerala and Bihar. A national average index may mask these gaps. In addition, global measures may sometimes ignore cultural diets and local food systems. The EAT-Lancet reference diet, though scientifically designed, was criticized for not fitting local traditions and being unaffordable for about 1.6 billion people worldwide¹⁰.

2.2 Rationale for Standardization

The idea of using a **standardized global nutritional index** comes from the need to make nutrition data **comparable, simple, and useful for decision-making** across different countries. Since malnutrition is a worldwide challenge that affects both rich and poor nations in different ways, having one common measure helps bring attention to the issue at the global level. For example, the **State of Food Security and Nutrition in the World (SOFI) 2024** report shows that about **2.6 billion people could not afford a healthy diet in 2022**, and this figure is only visible because

⁹ Global Hunger Index – <https://www.globalhungerindex.org>

¹⁰ Hirvonen et al., 2020, *Lancet Global Health* – [https://doi.org/10.1016/S2214-109X\(19\)30447-4](https://doi.org/10.1016/S2214-109X(19)30447-4)

countries report data in a standardized way.¹¹ Standardization makes it possible to compare Africa, Asia, and Latin America in one chart and to hold governments accountable to international goals like the Sustainable Development Goals (SDG 2: Zero Hunger)¹².

The advantages of standardization are clear. First, it allows for cross-country comparison, showing which countries are making progress and which are falling behind. Second, it enables benchmarking: policymakers can see how their country compares to similar nations and set goals accordingly. Third, it strengthens policy monitoring, since indicators are tracked in the same way over time. Fourth, it increases accountability, as global reports and rankings put public pressure on governments to act. Indices like the Global Hunger Index (GHI), despite their limitations, have been successful in drawing media attention and mobilizing debate on food insecurity (<https://www.globalhungerindex.org>).

At the same time, there are serious disadvantages. A standardized index can ignore local food cultures and may not fully reflect what healthy nutrition looks like in different places. For instance, diets in South Asia rely more on cereals and pulses, while in Latin America maize and beans dominate, and in Europe dairy and meat have higher roles. A single diet standard may not respect these variations. Another issue is socio-economic diversity. In large countries such as India, averages can hide inequality between states, districts, or between rural and urban areas. For example, while India shows improvement in child stunting nationally, some states still face very high rates¹³. A

¹¹ Fu, H., Moncayo, J. R., Bai, Y., Conti, V., Rissanen, M., & Cafiero, C. (2025, August 6). *Can everyone afford to eat healthy? New data show progress, but not everywhere*. World Bank Blogs. <https://blogs.worldbank.org/en/opendata/can-everyone-afford-to-eat-healthy--new-data-show-progress--but->

¹² (FAO et al., *State of Food Security and Nutrition in the World 2024*, <https://doi.org/10.4060/cc8520en>).

¹³ (Press Information Bureau. (2023, October 12). *Measurement of over 7 crore children shows wasting at about one-third of that used in Global Hunger Index*. Ministry of Women and Child Development, Government of India. <https://www.pib.gov.in/PressReleaseIframePage.aspx?PRID=1967164>).

single score can give a false sense of progress. Standardization can also lead to oversimplification—reducing complex realities into one number that may not capture important causes like sanitation, women’s education, or healthcare access.

In addition, critics have argued that some standardized indices are not well-matched to their stated goals. In India, for example, scientists and policymakers have pointed out that the GHI does not directly measure hunger, since it uses indicators like child mortality that are influenced by many factors beyond food intake.¹⁴ This raises the risk of misinterpretation and poor policy decisions if the index is taken at face value.

The rationale for standardization is strong when it comes to creating a shared language for global nutrition monitoring, enabling comparison, and driving international action. But it also has clear limits: without adaptation to local diets, socio-economic differences, and cultural practices, a standardized index may fail to guide effective solutions. A balanced approach is needed—one that uses standardized global measures for comparability but also integrates local indicators for accuracy and policy relevance.

3. Literature Review

The study of malnutrition has produced several global tools and indices that are widely used by international organizations, governments, and researchers. These tools aim to provide a clear picture of the scale and distribution of malnutrition and food insecurity. However, while they are useful for comparability, they also face strong critiques for not fully reflecting the social and cultural

¹⁴ (Singh, P., Kurpad, A. V., Verma, D., Nigam, A. K., Sachdev, H. S., Pandey, A., Hemalatha, R., Deb, S., Khanna, K., Awasthi, S., Toteja, G. S., Bansal, P. G., Gonmei, Z., & Bhargava, B., for the ICMR Expert Committee to Deliberate on Global Hunger Index. (2021). Global Hunger Index does not really measure hunger: An Indian perspective. *Indian Journal of Medical Research*, 154(3), 455–460. https://doi.org/10.4103/ijmr.ijmr_2057_21)

realities of different countries. This section reviews some of the most important existing indices, their strengths, their weaknesses, and what past case studies show about their effectiveness.

Global Hunger Index (GHI)

The Global Hunger Index (GHI) is one of the most visible and widely quoted measures of hunger. It combines four indicators: undernourishment, child wasting, child stunting, and child mortality (Global Hunger Index, <https://www.globalhungerindex.org>). The idea is to show both food access and child health outcomes in a single score, which is then used to rank countries. The GHI has been effective in drawing global attention to hunger, and the annual rankings receive wide media coverage. However, the GHI has been strongly criticized for being conceptually weak. In India, for example, experts argue that indicators like child mortality are influenced by many factors beyond food access, including sanitation, maternal education, and healthcare. The Government of India has also questioned the index for not capturing the country's actual food security achievements. This example shows that while the GHI is useful for advocacy, it may not always provide accurate or fair measures for specific countries.

Singh et al. (2021), writing on behalf of the ICMR Expert Committee, argue that the Global Hunger Index (GHI) is a misleading measure of hunger because its four indicators—undernourishment, stunting, wasting, and child mortality—do not directly capture food deprivation. The study shows that even in wealthier households with sufficient food access, high rates of stunting and wasting persist, which indicates that these conditions are also influenced by genetics, sanitation, and health factors rather than hunger alone. The authors further note that GHI gives excessive weight to under-five child outcomes, exaggerating the picture of hunger in India. They recommend developing country-specific methodologies and using tools like the Food Insecurity Experience Scale (FIES) for more accurate measurement. This critique highlights the limitations of standardized global

indices and supports the argument for hybrid frameworks that balance international comparability with local relevance.¹⁵

Deaton and Drèze (2009) highlight that in India, calorie intake has declined despite rising incomes, challenging the assumption that economic growth automatically improves nutrition. Their study shows that falling calorie consumption is linked not only to food supply but also to reduced energy requirements, improved health, sanitation, and dietary diversification. They emphasize that relying on income or calorie-based measures is misleading, as these overlook crucial determinants of nutrition. The paper also documents large disparities across states, rural and urban areas, and social groups, which national or global averages fail to capture. This evidence demonstrates that standardized global indices can oversimplify complex, context-specific nutrition realities, underscoring the need for locally adapted indicators¹⁶.

Hirvonen et al. (2020) show that the EAT–Lancet reference diet—often used as a global benchmark—had a median cost of US\$2.84/day (2011 international dollars) and was unaffordable for at least 1.58 billion people, mainly in South Asia and Sub-Saharan Africa. They also find the EAT–Lancet diet is 60% more expensive than the least-cost, nutrient-adequate diet, with fruits/vegetables and animal-source foods driving costs. Affordability as a share of mean per-capita household income rose from 6% in high-income to 89% in low-income countries, underscoring steep income and price constraints. These results suggest that using a single global diet standard inside

¹⁵ Singh, P., Kurpad, A. V., Verma, D., Nigam, A. K., Sachdev, H. S., Pandey, A., Hemalatha, R., Deb, S., Khanna, K., Awasthi, S., Toteja, G. S., Bansal, P. G., Gonmei, Z., & Bhargava, B., for the ICMR Expert Committee to Deliberate on Global Hunger Index. (2021). Global Hunger Index does not really measure hunger: An Indian perspective. *Indian Journal of Medical Research*, 154(3), 455–460. https://doi.org/10.4103/ijmr.ijmr_2057_21).

¹⁶ (Deaton, A., & Drèze, J. (2009). Food and nutrition in India: Facts and interpretations. *Economic and Political Weekly*, 44(7), 42–65. https://www.princeton.edu/~deaton/downloads/Food_and_Nutrition_in_India_Facts_and_Interpretations.pdf).

standardized indices can overstate policy feasibility where purchasing power is low, supporting your argument for hybrid indices that pair global comparability with local affordability modules.¹⁷

Raghunathan, Headey, and Herforth (2021) highlight that in rural India, nutritious diets remain largely unaffordable despite rising wages, with the Cost of a Recommended Diet (CoRD) consuming about 80–90% of women’s daily wages and 50–60% of men’s. Their estimates suggest that 63–76% of the rural poor could not afford a diet aligned with India’s national dietary guidelines in 2011. Dairy and fruits emerged as the most expensive food groups, while cereals remained the cheapest, showing how structural issues in India’s agriculture bias diets toward staples. The study argues for revised social protection policies, diversification of agriculture toward nutrient-rich foods, and improved monitoring tools to track affordability (Raghunathan et al., 2021). This underscores the importance of integrating local affordability metrics into any standardized global nutritional index¹⁸.

Food Insecurity Experience Scale (FIES)

The Food Insecurity Experience Scale (FIES), developed by the FAO, is another widely used tool. Unlike the GHI, it is based on household surveys where people answer standardized questions about their experiences with food insecurity—such as worrying about running out of food, skipping meals, or going hungry.¹⁹ Its strength lies in capturing the subjective experience of food insecurity, which makes it more people-centered. It also allows for disaggregated data at household level, which can be useful for national policies. However, the FIES has cross-cultural limitations. Some survey questions may not carry the same meaning in different cultural settings, and translation

¹⁷ (Hirvonen, K., Bai, Y., Headey, D., & Masters, W. A. (2020). Affordability of the EAT–Lancet reference diet: A global analysis. *The Lancet Global Health*, 8(1), e59–e66. [https://doi.org/10.1016/S2214-109X\(19\)30447-4](https://doi.org/10.1016/S2214-109X(19)30447-4))

¹⁸ Raghunathan, K., Headey, D., & Herforth, A. (2021). Affordability of nutritious diets in rural India. *Food Policy*, 99, 101982. <https://doi.org/10.1016/j.foodpol.2020.101982>

¹⁹ (FAO, FIES overview: <https://www.fao.org/in-action/voices-of-the-hungry>).

challenges can reduce accuracy . For example, rural households in Sub-Saharan Africa may interpret “skipping meals” differently from households in urban Latin America. This raises doubts about whether one standardized set of questions can fully capture food insecurity across very different societies.

Human Development Index (HDI) Nutrition Components

The **Human Development Index (HDI)**, created by UNDP, is not a nutrition-specific tool but includes health and education dimensions that overlap with nutrition. For example, life expectancy is partly shaped by nutrition outcomes. In some adaptations, nutrition indicators such as child stunting are used in **multidimensional poverty indices**²⁰. The HDI has been praised for broadening the view of development beyond GDP, but it faces the same criticism as other composite indices: it produces a single number that can mask inequality within countries. In India, the national MPI has shown that while some states like Kerala have strong nutrition outcomes, others like Bihar and Uttar Pradesh lag behind, even though both contribute to the same national average. This demonstrates why local disaggregation is necessary when using standardized measures.

Global Burden of Disease (GBD) Nutrition Metrics

The **Global Burden of Disease (GBD) study**, coordinated by the Institute for Health Metrics and Evaluation (IHME), uses nutrition-related risk factors to estimate the disease burden across countries. These include diet low in fruits, vegetables, whole grains, or iron, as well as risks from obesity and high blood sugar. The strength of GBD metrics is that they link nutrition directly to health outcomes and provide detailed cause-specific mortality and morbidity data. For example, the GBD 2019 report showed that poor diet is one of the leading causes of global disease burden. However, GBD methods are often **data-heavy and model-based**, meaning that in low-income

²⁰ (NITI Aayog, *National MPI 2023*, <https://www.niti.gov.in>).

countries with weak data systems, the results depend heavily on statistical estimation rather than real observations. This limits their reliability for local policy planning.²¹

Case Studies: South Asia, Sub-Saharan Africa, Latin America

Research from South Asia highlights the complexity of malnutrition measurement. India, despite economic growth, continues to face high rates of child stunting and anemia. Studies by Deaton and Drèze (2009, *EPW*) showed the paradox of declining calorie consumption despite rising incomes, suggesting that non-food factors like sanitation and disease play a central role. This indicates that indices based only on food availability may miss the bigger picture.²²

In Sub-Saharan Africa, the FIES and GHI have been used widely to compare hunger across countries. They highlight severe food insecurity, but critics argue that they ignore the resilience of local food systems and cultural diets. For instance, maize-based diets in parts of Africa may meet calorie needs but fail in micronutrients, yet global indices may not fully capture this gap.²³

In Latin America, where obesity and overweight are rising alongside undernutrition, global indices often struggle to capture the double burden of malnutrition. The Global Nutrition Report 2021 noted that countries like Mexico and Brazil face both child stunting and high adult obesity,

²¹ (Institute for Health Metrics and Evaluation. (n.d.). *Global burden of disease (GBD)*. University of Washington. <https://www.healthdata.org/research-analysis/gbd>)

²² ((Deaton, A., & Drèze, J. (2009). Food and nutrition in India: Facts and interpretations. *Economic and Political Weekly*, 44(7), 42–65. https://www.princeton.edu/~deaton/downloads/Food_and_Nutrition_in_India_Facts_and_Interpretations.pdf.)

²³ (Global Nutrition Report. (2021). Latin America and the Caribbean: Nutrition profile. Global Nutrition Report. <https://globalnutritionreport.org/resources/nutrition-profiles/latin-america-and-caribbean/>)

showing that traditional hunger measures are no longer sufficient. This supports the idea that a more flexible or multidimensional index is needed.²⁴

Research Gaps

The review of these indices shows three major gaps. First, limited cultural sensitivity: global indices rarely adapt to local diets or body types (for example, WHO's Asian BMI cut-offs show why standard thresholds may misclassify risk). Second, data gaps and quality issues: many countries rely on outdated surveys or modeled data, which weakens reliability. Third, urban–rural and regional differences: national averages hide inequalities, especially in large countries like India or Nigeria. These gaps show why a purely standardized global index cannot be enough on its own.

Table 1. Comparison of Key Global Nutrition Indices

Index / Tool	Main Indicators Used	Strengths	Weaknesses / Critiques	Example / Use Case
Global Hunger Index (GHI)	Undernourishment, child wasting, child stunting, child mortality	Simple to communicate, global rankings gain media and policy attention	Does not directly measure hunger; child mortality influenced by non- food factors; criticized in India for poor fit	Used to rank >100 countries each year; India's ranking sparked major policy

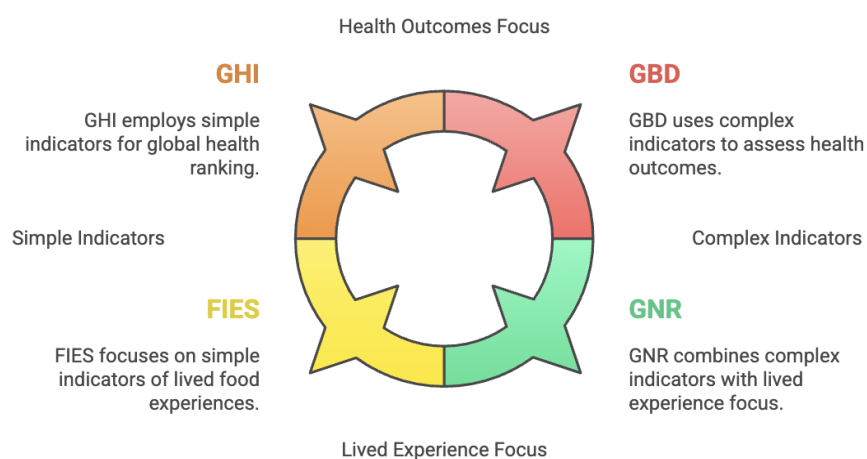
²⁴ (Global Nutrition Report. (2021). A world free from malnutrition: An assessment of progress towards the global nutrition targets. Global Nutrition Report.

https://media.globalnutritionreport.org/documents/Chapter_1_2021_Global_Nutrition_Report.pdf

				debate
Food Insecurity Experience Scale (FIES)	Household survey questions on food access, skipping meals, worrying about food	Captures people's lived experiences; allows disaggregated analysis at household level	Cultural/translation issues may change meaning; may not reflect nutrition quality	FAO reports global food insecurity prevalence (SOFI reports)
Human Development Index (HDI) – Nutrition Components / Multidimensional Poverty Index (MPI)	Life expectancy, education, income; sometimes includes stunting/underweight	Links nutrition to wider human development; disaggregation possible	Averages hide inequality between states, rural/urban, or social groups	NITI Aayog's MPI in India includes nutrition and shows state-level differences
Global Burden of Disease (GBD) Nutrition Metrics	Dietary risk factors (low fruit/veg, high sugar, obesity, anemia); linked to	Links diet to health outcomes (mortality,	Data-heavy; estimates rely on models in low-data settings; less	GBD 2019 showed poor diet as top global

	disease outcomes	DALYs); global comparability	usable for local policy	health risk factor
Global Nutrition Report Indicators	Diet diversity, affordability, overweight, undernutrition, anemia	Combines diet, affordability, and outcomes in one profile; widely cited	Still aggregate- level; doesn't solve cultural/local diet diversity issues	2021 Report highlighted double burden of malnutrition in Latin America

Comparative Analysis of Global Health and Nutrition Indices



4. Methodological Considerations

Global nutrition measures draw on many data sources, each with strengths and limits.

Countries report supply-side food data through FAO Food Balance Sheets (FBS), which estimate

per-capita availability of calories and nutrients from production, trade, stocks, and non-food uses; these are standardized but do not show who actually eats what at household level.²⁵ Household and individual surveys add demand-side detail. The Demographic and Health Surveys (DHS) measure child and adult nutrition (stunting, wasting, underweight/overweight) and infant and young child feeding, using harmonized protocols across countries.²⁶ The Food Insecurity Experience Scale (FIES) captures people's lived difficulties in accessing food through eight standard questions and a common analytic model to classify severity²⁷. For global child indicators, the UNICEF-WHO-World Bank Joint Malnutrition Estimates (JME) team applies a transparent method to harmonize and update national stunting, wasting, and overweight time series, improving cross-country comparability.²⁸ To connect diets to disease, the Global Burden of Disease (GBD) program models health loss from 15 diet-related risks, offering comparable risk estimates over time and place.²⁹ Finally, affordability metrics—central to many indices—use the Cost and Affordability of a Healthy Diet (CoAHD),

²⁵ (Food and Agriculture Organization of the United Nations. (2001). Food balance sheets: Applications and uses – I. Standardization of food balance sheets. FAO. <https://www.fao.org/fileadmin/templates/ess/documents/methodology/fbs3-edg.pdf>)

²⁶ (The DHS Program. (n.d.). Nutrition indicator data. ICF. <https://dhsprogram.com/topics/Nutrition/index.cfm>)

²⁷ (Food and Agriculture Organization of the United Nations. (n.d.). Voices of the Hungry: Food Insecurity Experience Scale (FIES). FAO. <https://www.fao.org/in-action/voices-of-the-hungry/fies/en/>).

²⁸ UNICEF, World Health Organization, & The World Bank. (n.d.). *Joint child malnutrition estimates (JME)*. World Health Organization. Retrieved September 25, 2025, from <https://www.who.int/teams/nutrition-and-food-safety/monitoring-nutritional-status-and-food-safety-and-events/joint-child-malnutrition-estimates>

²⁹ Institute for Health Metrics and Evaluation. (n.d.). *Diet*. HealthData.org. Retrieved September 25, 2025, from <https://www.healthdata.org/research-analysis/health-topics/diet>

which prices the cheapest local foods that meet healthy-diet standards and converts them to a common currency with PPP from the World Bank's International Comparison Program (ICP)³⁰.

These sources face measurement problems that affect any composite index. Underreporting of foods eaten away from home, recall bias in 24-hour recalls or food-frequency tools, and seasonality of diets in low- and middle-income countries can distort intake estimates; seafood, wild foods, and informal-market purchases are often missed.³¹ FBS, while consistent, infer availability rather than intake and can undercount processed derivatives if commodity transformations are not well tracked.³² Anthropometry can suffer from measurement error (e.g., height heaping), while anemia estimates vary with device and adjustment choices; JME methods help, but residual uncertainty remains.³³ Affordability trends can shift when PPP benchmarks are revised (e.g., 2021 ICP update), so indices should document base years and re-estimate time series to keep comparisons fair³⁴.

³⁰ Food and Agriculture Organization of the United Nations. (2025, July). *The cost and affordability of a healthy diet (CoAHD) indicators: Methods and data sources*. FAO. https://files-faostat.fao.org/production/CAHD/Methods_Brief_FAOSTAT_CoAHD_indicators.pdf

³¹ Padula de Quadros, V., Balcerzak, A., Allemand, P., de Sousa, R. F., Bevere, T., Arsenault, J., Deitchler, M., & Holmes, B. A. (2022). Global trends in the availability of dietary data in low- and middle-income countries. *Nutrients*, 14(14), 2880. <https://doi.org/10.3390/nu14142880>

³² Food and Agriculture Organization of the United Nations. (n.d.). *Food balance sheets: Overview, methodology and country support* [Presentation]. SESRIC / FAO Webinar on “Enhancing Food Security in OIC Countries: An Overview of the Food Balance Sheet.” https://www.sesric.org/imgs/news/2429_PRESENTATION_FAO_FBS-OVERVIEW.pdf

³³ (UNICEF, World Health Organization, & The World Bank. (n.d.). Joint child malnutrition estimates (JME). World Health Organization. Retrieved September 25, 2025, from <https://www.who.int/teams/nutrition-and-food-safety/monitoring-nutritional-status-and-food-safety-and-events/joint-child-malnutrition-estimates>)

³⁴ World Bank. (2024, May 30). *Global purchasing power parities data released for 2021* [Press release]. <https://www.worldbank.org/en/news/press-release/2024/05/30/global-purchasing-power-parities-data-released-for-2021>

Building a composite index requires clear rules: choose indicators (e.g., affordability, diet diversity, stunting, anemia), normalize them to a common scale (z-scores or min–max), weight them (equal weights, expert weights, or data-driven methods like PCA), and aggregate (often a linear or geometric mean). Each choice changes results, so authors should publish a methods annex and run sensitivity checks (e.g., reweighting, dropping one indicator at a time). To protect reliability (consistent results) and validity (measuring what matters), indices should: (1) use standard definitions (JME for child outcomes, FIES for experience-based food insecurity, CoAHD for affordability), (2) show disaggregated results (sex, age, rural/urban, subnational), and (3) flag data quality (survey year, sampling, season). This approach keeps cross-country comparability while being honest about limits—especially where data are modeled (GBD) or supply-based (FBS). In short, a useful global nutritional index is as much about transparent methods and quality control as it is about the indicators themselves³⁵.

5. Effectiveness of a Standardized Global Index

A standardized global nutritional index can be a powerful tool—but its value depends on how it is built and how it is used. In this section, we assess how effective a single, comparable index is for measuring and addressing malnutrition across very different countries and cultures. We look at the main strengths (why a global index helps) and the main weaknesses (where it can mislead), using current, reliable evidence.

³⁵ Food and Agriculture Organization of the United Nations. (2024, July 24). *Healthy diets remain unaffordable for a third of the world's population: SOFI 2024 report highlights a cost-hunger trap plaguing some regions in the wake of the COVID-19 pandemic*. FAO. <https://www.fao.org/newsroom/detail/healthy-diets-remain-unaffordable-for-a-third-of-the-world-s-population/en>

5.1 Strengths

Comparability enables accountability.

A single, shared frame lets governments, media, and citizens see where their country stands and how it is changing over time. For example, the UN system’s annual *State of Food Security and Nutrition in the World (SOFI)* assembles standardized indicators on hunger, diet affordability, and malnutrition. Because countries report in a common way, the world can track broad progress toward SDG 2 (Zero Hunger) and see which regions are lagging. This comparability matters for public debate and policy pressure. In 2024, SOFI reported new affordability estimates and continued to monitor hunger and diet quality using harmonized methods—

A global index can help leaders spot priorities fast. When standardized data show that about 2.6 billion people cannot afford a healthy diet and the global average cost of a healthy diet is about \$4.46 (PPP) per person per day, it signals that food prices, purchasing power, and social protection deserve immediate attention. This kind of top-line number is simple, memorable, and focuses policy on price dynamics, wages, targeted transfers, and school meals. It also supports cross-country learning: countries that shielded poor households from food inflation and expanded school feeding saw faster gains³⁶. Clear, comparable indicators help governments and partners align resources. The Scaling Up Nutrition (SUN) Movement uses a formal monitoring, evaluation, accountability, and learning (MEAL) system and an accountability framework to track commitments and progress. Donors and governments rely on such standardized tracking to plan budgets, check delivery, and fix

³⁶ (Food and Agriculture Organization of the United Nations, International Fund for Agricultural Development, United Nations Children’s Fund, World Food Programme, & World Health Organization. (2024). The state of food security and nutrition in the world 2024: Financing to end hunger, food insecurity and malnutrition in all its forms. FAO.

<https://www.who.int/publications/m/item/the-state-of-food-security-and-nutrition-in-the-world-2024>)

gaps in nutrition programming. Without common indicators and templates, joint planning and pooled funds would be much harder³⁷.

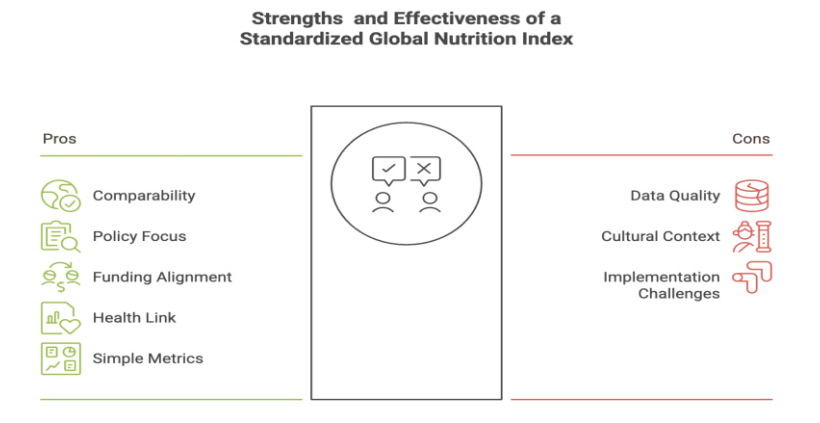
Global platforms like the **Global Burden of Disease (GBD)** translate diet patterns into health risks and outcomes (mortality, DALYs) using the same definitions everywhere. That lets countries compare how much disease is due to poor diets and which dietary risks matter most (for example, low whole grains, low fruits, high sodium), informing cost-effective prevention and treatment³⁸.

A single number or compact dashboard (“how affordable is a healthy diet?”; “what share of children are stunted?”) cuts through complex datasets. This helps journalists, civil society, and parliaments to keep nutrition visible—especially when many other issues compete for attention. SOFI, the Global Nutrition Report, and similar scorecards succeed largely because standardization makes their messages simple to use³⁹.

³⁷ (Food and Agriculture Organization of the United Nations, International Fund for Agricultural Development, United Nations Children’s Fund, World Food Programme, & World Health Organization. (2024). The state of food security and nutrition in the world 2024: Financing to end hunger, food insecurity and malnutrition in all its forms. FAO. <https://www.who.int/publications/m/item/the-state-of-food-security-and-nutrition-in-the-world-2024>) . [World Bank Blogs+1](#)

³⁸ Scaling Up Nutrition. (n.d.). Monitoring, evaluation, accountability and learning (MEAL) system. Scaling Up Nutrition Movement. <https://scalingupnutrition.org/about/how-we-do-it/monitoring-ev>

³⁹ Food and Agriculture Organization of the United Nations, International Fund for Agricultural Development, United Nations Children’s Fund, World Food Programme, & World Health Organization. (2024). The state of food security and nutrition in the world 2024: Financing to end hunger, food insecurity and malnutrition in all its forms. FAO. https://data.unicef.org/wp-content/uploads/2024/07/SOFI2024_Report_EN_web.pdf



5.2 Weaknesses

Cultural bias and diet misfit. A universal index can carry hidden assumptions about “good diets.” Global diet models (like EAT-Lancet) were designed from first principles, but real diets are local. In many places, staple-heavy or plant-forward patterns can be healthy and culturally appropriate, yet a universal “reference basket” may not fit prices, preferences, or cooking practices. Studies have shown that the reference healthy diet can be unaffordable for large populations, underscoring the risk of using a one-size basket to judge countries or to set unattainable goals⁴⁰.

Malnutrition looks different across settings. India’s main challenge remains child stunting and anemia in many districts, while the United States struggles with obesity and diet-related NCDs; Latin America faces a double burden (child undernutrition alongside adult overweight). A single composite score may not capture these different policy priorities and can blur the real problems that need attention in each place. Region profiles and research show how overweight is rising even where

⁴⁰ Fu, H., Rosero Moncayo, J., Bai, Y., Conti, V., Rissanen, M., & Cafiero, C. (2025, August 6). Can everyone afford to eat healthy? New data show progress, but not everywhere. World Bank Blogs.

<https://blogs.worldbank.org/en/opendata/can-everyone-afford-to-eat-healthy--new-data-show-progress--but->

stunting persists, especially in Latin America and parts of urban Africa—evidence that a one-number view can hide the double burden⁴¹.

National averages can hide within-country gaps—by state, district, ethnicity, income, or rural-urban status. India’s Multidimensional Poverty Index (MPI) demonstrates how sub-national breakdowns change the story: some states have strong outcomes, others lag far behind, even though the national average improves. A global index that reports only the average can create a false sense of progress and weaken the focus on the poorest groups (tribal communities, refugees, urban informal workers). Good practice is to pair any global score with disaggregated results⁴².

The Global Hunger Index (GHI) combines undernourishment, child wasting, child stunting, and child mortality. Critics, including India’s public health community, argue that this mix does not directly measure “hunger” and is influenced by non-food determinants like sanitation and health services. When such an index is taken at face value, it can misdirect debate and policy. The key lesson is to ensure construct validity: the index should measure what it claims to measure, or else be labeled clearly as a broader nutrition/health risk index rather than “hunger.”⁴³

Health risks at a given BMI differ by population. The WHO Expert Consultation (2004) found that Asian populations face higher metabolic risk at lower BMI than the global standard (≥ 25 kg/m²). If a global index uses one universal BMI cut-off, it may underestimate risk in Asian groups

⁴¹ Global Nutrition Report. (n.d.). Country nutrition profiles: Latin America and Caribbean. Development Initiatives. <https://globalnutritionreport.org/resources/nutrition-profiles/latin-america-and-caribbean/>

⁴² (NITI Aayog. (2023). National multidimensional poverty index: A progress review 2023. Government of India. <https://www.niti.gov.in/sites/default/files/2023-08/India-National-Multidimensional-Poverty-Index-2023.pdf>).

⁴³ Singh, P., Kurpad, A. V., Verma, D., Nigam, A. K., Sachdev, H. S., Pandey, A., Hemalatha, R., Deb, S., Khanna, K., Awasthi, S., Toteja, G. S., Bansal, P. G., Gonmei, Z., & Bhargava, B. (2021). Global Hunger Index does not really measure hunger – An Indian perspective. *Indian Journal of Medical Research*, 154(4), 559–564. https://doi.org/10.4103/ijmr.ijmr_2057_21

and overestimate in others. This argues for context-specific cut-offs (or additional measures like waist circumference) inside a common global frame⁴⁴.

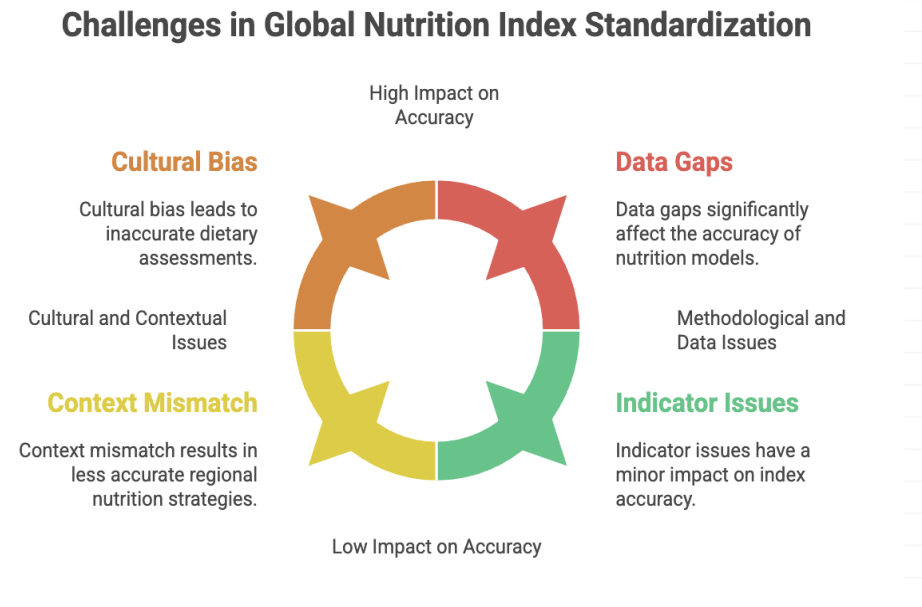
Some global metrics rely on modeled estimates where surveys are scarce or outdated. For instance, GBD uses complex models to fill data gaps; FAO Food Balance Sheets infer availability rather than actual intake; household diet surveys can miss informal markets, wild foods, or meals eaten away from home. These methods are valuable, but they introduce uncertainty. If a global index does not show data quality flags and confidence ranges, users may read too much into small rank changes. (This is one reason many agencies now publish technical annexes and uncertainty intervals.

When donors tie funds to a narrow set of indicators, countries may focus on improving the score rather than the underlying systems (water and sanitation, primary healthcare, women's education, food environments). The SUN Movement tries to reduce this risk by tracking actions and finance alongside outcomes, but the danger remains if the global index becomes the only yardstick. The fix is to combine a global core with context-specific modules and to track inputs and systems as well as outcomes.

A standardized global index is most effective for monitoring, international comparison, agenda-setting, and resource mobilization. It gives a shared language, sharper headlines, and clearer global pictures—vital for SDG tracking and for making the economic case to invest in nutrition. But to address malnutrition, countries need a hybrid approach: keep a global core (for comparability) and add local modules (for cultural diets, price realities, sub-national gaps, and population-specific risk thresholds). Pair the index with disaggregation, uncertainty ranges, and transparent methods, and use it alongside program indicators (coverage, quality, financing). With these safeguards, a standardized

⁴⁴ WHO Expert Consultation. (2004). Appropriate body-mass index for Asian populations and its implications for policy and intervention strategies. *The Lancet*, 363(9403), 157–163. [https://doi.org/10.1016/S0140-6736\(03\)15268-3](https://doi.org/10.1016/S0140-6736(03)15268-3)

global index becomes a reliable compass for the journey—while the detailed local maps guide the actual route.



6. Alternatives and Improvements

While a standardized global nutritional index has clear strengths, it is not enough by itself to guide effective action. To better measure and address malnutrition, several alternatives and improvements have been suggested in research and policy debates. These include **context-sensitive approaches**, **mixed methods that combine numbers with local voices**, **the use of technology and big data**, and a **hybrid framework** that combines global comparability with local flexibility.

6.1 Context-Sensitive Approaches

One important improvement is to make indices more region-specific. For example, the African Union’s Continental Nutrition Accountability Scorecard presents each country’s progress on child stunting, wasting, and exclusive breastfeeding, but allows African governments to compare themselves to regional peers rather than only to global averages (African Union, 2022). This makes

the results more meaningful for local policymakers, as they can set goals in line with their region's realities.

In South Asia, indicators like anemia prevalence and minimum dietary diversity for women are especially relevant, given cultural diets and the high burden of micronutrient deficiencies. Similarly, in Latin America, indices that include overweight and obesity reflect the region's growing "double burden of malnutrition." By adapting the basket of indicators to local diets and challenges, indices become more accurate and useful.

Another way to improve sensitivity is to include dietary diversity indicators. Research shows that national calorie supply is no longer enough to judge nutrition progress. For example, studies in India demonstrate that households may meet energy needs but fail in diet quality, with low intake of protein and micronutrients (Deaton & Drèze, 2009, *EPW*). Including measures like the **Dietary Diversity Score** or the **Food Variety Score** ensures that indices reflect not just how much people eat, but also how balanced their diets are.

6.2 Mixed Methods

Numbers alone cannot tell the full story. A second improvement is to combine quantitative data (indices, surveys, composite scores) with qualitative community insights. For example, FAO and UNICEF have piloted participatory food security assessments, where communities discuss their own definitions of food adequacy and health. These methods capture local knowledge about food cultures, seasonal changes, and barriers to healthy diets that are often invisible in global statistics (FAO, "Voices of the Hungry" project).

Qualitative approaches can also help policymakers understand why nutrition outcomes differ between groups. For example, in rural India, women may prioritize feeding children and men first, leading to unequal nutrient distribution within households. A standardized index would show only

child stunting rates, but interviews and focus groups reveal the underlying cultural practice. By combining the two, policymakers can design culturally sensitive interventions, such as nutrition education or women's empowerment programs.

6.3 Role of Technology and Big Data

New technologies offer promising ways to improve measurement. Mobile phone surveys have become a cost-effective method for collecting real-time food security and diet data, especially during crises like the COVID-19 pandemic. For example, the World Bank's High Frequency Phone Surveys provided rapid updates on food access during lockdowns.

Satellite imaging can also help. Crop diversity and land-use data from satellites can be linked to dietary diversity at household level, giving a new way to track the availability of nutritious foods across regions. Similarly, machine learning models can predict local food insecurity "hot spots" by combining satellite data on vegetation, rainfall, and markets.

Another frontier is AI-driven dietary assessment. Tools that scan food purchases, photos of meals, or online price data can estimate dietary quality with much more detail than traditional surveys. These innovations can help overcome problems of underreporting, recall bias, and data delays that affect existing indices. However, they require careful design to respect privacy and to ensure data is accessible in low-income settings.

6.4 Hybrid Framework Proposal

Perhaps the most practical solution is to adopt a hybrid framework that combines the strengths of both standardized and local approaches. In this model, a global core index would use a small set of common indicators—such as the Cost and Affordability of a Healthy Diet (CoAHD), child stunting and wasting (JME data), and household food insecurity (FIES). These indicators provide global comparability and accountability.

Alongside this, countries would be encouraged to add local modules. For India, this might include anemia prevalence, minimum dietary diversity, and BMI cut-offs adjusted for Asian populations (WHO, 2004). For African countries, local modules might track cassava or maize dependence and micronutrient fortification coverage. For Latin America, local modules could include obesity prevalence and consumption of ultra-processed foods.

This approach balances global visibility with cultural relevance. It also allows countries to address their unique challenges while still contributing to the global picture. The National Multidimensional Poverty Index (MPI) in India provides a good example: it uses a common framework but adapts indicators to local realities, making it both comparable and actionable.

7. Case Studies

India: one country, many realities

India shows why a single national score can hide big gaps. NFHS-5 (2019–21) reports slow improvement in child stunting and persistent high anemia among women and children, but the averages mask sharp differences between states, districts, rural and urban areas. India's National Multidimensional Poverty Index (MPI), led by NITI Aayog, includes nutrition as a core health indicator and presents disaggregation by state and region; this makes inequalities visible and helps target action. India's POSHAN Abhiyaan adds a program lens—real-time dashboards, convergence across departments, and a focus on the first 1,000 days of life—to push coverage and behavior change⁴⁵. Together, these sources show that while a **standardized global index** is useful for

⁴⁵ NITI Aayog. (2019). *Transforming nutrition in India: POSHAN Abhiyaan, first progress report, December 2018*. Government of India. https://www.niti.gov.in/sites/default/files/2020-02/POSHAN_Abhiyaan_first_progress_report_6_Feb_2019.pdf

signaling risk, India needs **subnational tracking** to guide spending and delivery⁴⁶. The debate around the **Global Hunger Index (GHI)** also illustrates the risk of misfit: Indian public health bodies argue that GHI's indicators (e.g., child mortality) do not directly measure “hunger,” and the government has formally questioned its methodology. The policy lesson is clear: pair any global index with **NFHS and MPI detail** and program dashboards like **POSHAN** to plan action at the ground level⁴⁷.

Brazil: Zero Hunger, cash transfers, and local food systems

Brazil's Fome Zero (Zero Hunger) strategy and Bolsa Família cash transfers are often cited as a successful package: higher household income, school meals using local procurement, nutrition education, and stronger primary health care. FAO's evaluation shows how a joined-up policy mix— income support plus food and health programs—reduced extreme poverty and improved food access; later reviews describe how Brazil's approach informed global debates on social protection and food security. Importantly, results were not only about more calories; the program also tried to link family farmers to school feeding markets, strengthening local food systems and dietary diversity. For a global index, Brazil's experience says: a single score can attract attention, but multisector programs and local supply chains do the real work. When fiscal or political shifts weaken these programs,

⁴⁶ (Press Information Bureau. (2021, December 3). Global Hunger Index 2021 [Press release]. Ministry of Women and Child Development, Government of India

⁴⁷ Press Information Bureau. (2021, December 3). Global Hunger Index 2021. Ministry of Women and Child Development, Government of India. <https://www.pib.gov.in/PressReleasePage.aspx?PRID=1777719> (<https://www.pib.gov.in/PressReleasePage.aspx?PRID=1777719>) . [Press Information Bureau+6DHS Program+6NITI Aayog+6](#)

gains can slow down; another caution that sustained financing and institutional stability matter as much as measurement⁴⁸.

Ethiopia: stunting reduction and the limits of one number

Ethiopia has pursued stunting reduction through the Seqota Declaration, a multi-sector plan to eliminate stunting by 2030. Government and partners have focused on water and sanitation, primary health care, nutrition services, and food systems, with clear coverage targets and a roadmap for scale-up. Historical data show significant declines in stunting over the last three decades, though progress varies by region and by rural–urban location⁴⁹. Recent UNICEF–WFP partnership reports highlight system-strengthening and the push to reach high-burden districts. For a standardized global index, Ethiopia’s case shows both strength and limit: the index can track national progress and support donor coordination, but **program managers need more granular indicators** (annual coverage, quality of services, local food prices) to manage performance. Without subnational detail, one national number can hide the places where stunting remains very high and where **bottlenecks**—like safe water or health worker density—block change⁵⁰.

⁴⁸ da Silva, J. G., Del Grossi, M. E., & de França, C. G. (Eds.). (2011). *The Fome Zero (Zero Hunger) program: The Brazilian experience*. Food and Agriculture Organization of the United Nations. <https://www.fao.org/4/i3023e/i3023e.pdf>

⁴⁹ Ministry of Health, Ethiopia. (2015). Seqota Declaration. Government of Ethiopia. https://www.moh.gov.et/en/Seqota_Declaration?language_content_entity=en

⁵⁰ Bedingar, E. (2019). *The Seqota Declaration: Demonstrating intersectoral collaboration* (with excerpts from A. Bach, E. Gregor, & S. Sridhar; reviewed by K. Admasu). Harvard Ministerial Leadership Program. <https://ministerialleadership.harvard.edu/wp-content/uploads/2025/02/2019-The-Seqota-Declaration-Demonstrating-Intersectoral-Collaboration-FINAL.pdf>

United States: obesity, food deserts, and the double burden

The United States shows that malnutrition is not only about poverty or undernutrition. National data point to high and rising obesity, including severe obesity in adults, alongside pockets of food insecurity. The USDA's Food Access Research Atlas maps "low-income, low-access" areas, often called food deserts, where many people live far from full-service grocery stores and face transport barriers. At the same time, USDA food security statistics show millions of households with low food security, even in a high-income country. A global index that focuses only on undernourishment would miss this reality. The U.S. case argues for domain-specific modules inside any global framework: one set for diet-related NCD risk (obesity, diabetes, high sodium/sugar intakes), and another for food access (distance, affordability, transport). It also shows why local policy tools such as zoning for healthy retail, incentives for fresh food in underserved areas, and nutrition standards in schools, are needed alongside national monitoring.

Across these four settings, a pattern appears. A standardized global index is valuable for visibility, comparison, and funding conversations; it helps tell the world where hunger and poor diets remain.

But the action signal comes from context: India's subnational NFHS/MPI and POSHAN dashboards, Brazil's social protection plus local procurement, Ethiopia's service coverage and WASH, and the United States' food access maps and NCD focus. Two practical rules follow. First, always disaggregate (by place, age, sex, wealth, and social group) to avoid hiding the poorest. Second, pair the global score with local program indicators and equity lenses so budgets and delivery reach the groups that standard averages miss. In short, the case studies support a hybrid model: keep a global core for comparability, and add country-specific modules that fit diets, diseases, and systems on the ground.

8. Policy and Practical Implications

Global nutritional indices are valuable tools, but their impact depends on how governments and international organizations use them. The first principle is to avoid “**one-size-fits-all**” **approaches**. A global index can show big trends, highlight where countries stand, and mobilize action for **Sustainable Development Goal 2 (Zero Hunger)**. However, if policymakers rely only on a single number, they risk missing local differences and setting inappropriate policies.⁵¹

Governments should use global indices as a starting point for accountability and international comparison, not as the final measure of success. For example, India’s experience with the Global Hunger Index (GHI) shows that indicators must be cross-checked with national surveys like NFHS and NITI Aayog’s MPI to guide domestic policy. The same applies in Africa, where continental scorecards are most effective when combined with local dietary diversity and anemia data.

International organizations, donors, and partners can use standardized indices to prioritize funding and to monitor shared commitments. The Scaling Up Nutrition (SUN) Movement and the Global Nutrition Report already do this by using common indicators to track government and donor investments. But for maximum effectiveness, they should also require subnational disaggregation (rural/urban, gender, wealth groups) and encourage governments to build local modules that reflect their unique nutrition challenges.⁵²

Another key implication is to design layered monitoring systems. At the global level, agencies like FAO, WHO, and UNICEF can maintain a core set of standardized measures—such as

⁵¹ (Global Nutrition Report. (2020). Introduction: Towards global nutrition equity. In 2020 Global Nutrition Report: Action on equity to end malnutrition (Chapter 1). Development Initiatives. <https://globalnutritionreport.org/reports/2020-global-nutrition-report/introduction-towards-global-nutrition-equity/>)

⁵² (Coile, A., Wun, J., Kothari, M. T., Hemminger, C., Fracassi, P., & Di Dio, D. (2021). Scaling up nutrition through multisectoral planning: An exploratory review of 26 national nutrition plans. *Maternal & Child Nutrition*, 17(S2), e13171. <https://doi.org/10.1111/mcn.13171>)

the Cost of a Healthy Diet, child stunting and wasting, and food insecurity experience. At the national level, governments can add context-specific indicators such as Asian BMI cut-offs (for South Asia), anemia and dietary diversity (for India), or overweight/obesity (for Latin America). This layered approach protects comparability across countries while ensuring relevance for local action⁵³.

Finally, policymakers should link indices directly to planning and budgeting cycles. Ethiopia's Seqota Declaration, Brazil's Zero Hunger program, and India's POSHAN Abhiyaan show that when nutrition data feeds into real-time dashboards, state budgets, and accountability scorecards, progress accelerates. International donors should also align financing with countries' local modules, not just with global averages. Thus we can say global indices are most effective when they are used as accountability signals at the international level, and when they are paired with local, disaggregated, and program-relevant measures for national action. A layered approach—global comparability plus local adaptability—is the responsible way forward.

9. Conclusion

The evidence shows that a standardized index is indeed useful for broad monitoring, comparison, and advocacy. Tools like the SOFI report, GHI, and Global Nutrition Report make nutrition visible at the global level, allow countries to benchmark progress, and attract funding. They provide a shared language for international cooperation and help track commitments under SDG 2.

At the same time, standardized indices have clear limitations. They can carry cultural bias by assuming one model of a healthy diet, overlook local food systems, and misclassify health risks

⁵³ Emmerich, S. D., Fryar, C. D., Stierman, B., & Ogden, C. L. (2024, September). *Obesity and severe obesity prevalence in adults: United States, August 2021–August 2023* (NCHS Data Brief No. 508). National Center for Health Statistics, Centers for Disease Control and Prevention.

when they apply uniform thresholds (such as global BMI cut-offs). They also depend on national averages, which often hide inequalities between states, districts, or social groups. The debate over India's GHI ranking, the double burden in Latin America, and the persistence of food deserts in the U.S. all show that one global score cannot capture the full reality.

The findings suggest that the best path is not to abandon global indices, but to strengthen them with hybrid frameworks. A global “core” index—based on affordability of a healthy diet, stunting/wasting data, and food insecurity experience—can provide comparability and accountability. Around this, each country should add localized modules that capture their specific nutrition challenges, whether it is anemia in India, maize dependence in Africa, or obesity in Latin America.

Future research should focus on testing hybrid models, developing better diet quality indicators, and using technology (AI, mobile surveys, satellite data) to close data gaps. More work is also needed on how to make indices equity-sensitive, so that they show not just national averages but also the situation of the poorest groups.

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